Analysis of Mechanical system :

There are of two types :

a) Translational system

b) Rotational system.

Translational : The mechanical system in which motion is taking place along a straight line are called Translational type.

These systems are characterized by "displacement" "linear velocity" and "linear acceleration".

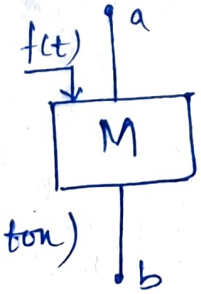
** The basic elements involved in mechanical system are :→

i) Mass ii) Spring iii) Damper/dashpot.

i) Mass (M)

The mass (M) is the interia elements.

lets assume that the mass (M) is subjected to force f(t).



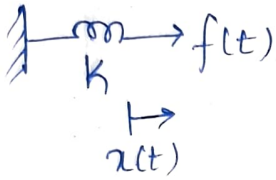
The reaction force f_m is equal to product of mass and acceleration and it is Opposite to the direction of the applied force

$$f_m = M \frac{d^2x(t)}{dt^2} \text{ (Newton)}$$

ie $\boxed{F_m(s) = Ms^2 X(s)}$

ii) Spring (K)

Force required to cause displacement '$x(t)$' in the spring is propotional to the force applied.

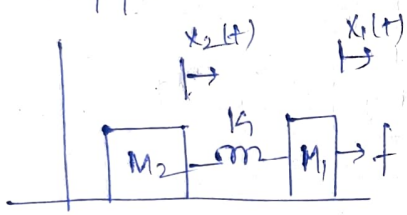


The force can be written as,

$$f(t) = K \cdot x(t)$$

$$L.T \Rightarrow \boxed{F(s) = K X(s)}$$

if the spring is connected between two moving elements having mases M_1 & M_2, Force is applied



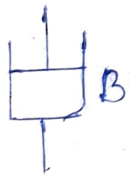
$$f(t) = K[x_1(t) - x_2(t)]$$

taking laplace transform.

$$F(s) = K[X_1(s) - X_2(s)]$$

iii) Damper (B)

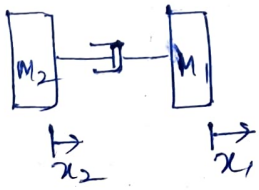
The network representation of damping action is a dashpot.



The reaction force f_B is equal to the product of damping coefficient 'B' and relative velocity of the two ends of the dashpots.

$$f_B(t) = B \cdot \frac{dx(t)}{dt} \xrightarrow{L.T} \boxed{F_B(s) = B s X(s)}$$

If the dashpot is connected in between two masses



then, $f(t) = B\left[\frac{dx_1(t)}{dt} - \frac{dx_2(t)}{dt}\right]$